we know something but not everything about their state, and we can only make predictions based on probability. Think of a balloon filled with air. I can measure it: measure its shape, its volume, its pressure, its temperature … But the molecules of air inside the balloon are moving rapidly within it, and I do not know the exact position of each of them. This prevents me from predicting with precision how the balloon will behave. For instance, if I untie the knot that seals it and let it go it will deflate noisily, rushing and colliding here and there in a way which is impossible for me to predict. Impossible, because I only know its shape, volume, pressure and temperature. The bumping about here and there of the balloon depends on the detail of the position of the molecules inside it, which I don't know. Yet even if I can't predict everything exactly, I can predict the probability that one thing or another will happen. It will be very improbable, for instance, that the balloon will fly out of the window, circle the lighthouse down there in the distance and then return to land on my hand, at the point where it was released. Some behaviour is more probable, other behaviour more improbable.

In this same sense, the probability that when molecules collide heat passes from the hotter bodies to those which are colder can be calculated, and turns out to be much greater than the probability of heat moving toward the hotter body.

The branch of science which clarifies these things is called statistical physics, and one of its triumphs, beginning with Boltzmann, has been to understand the probabilistic nature of heat and temperature, that is to say, thermodynamics.

At first glance, the idea that our ignorance implies something about the behaviour of the world seems irrational: the cold teaspoon heats up in hot tea and the balloon flies about when it is released regardless of what I know or don't know. What does what we know or don't know have to do with the laws that govern the world? The question is legitimate; the answer to it is subtle.

Teaspoon and balloon behave as they must, following the laws of physics in complete independence from what we know or don't know about them. The predictability or unpredictability of their behaviour does not pertain to their precise condition; it pertains to the limited set of their properties with which we interact. *This* set of properties depends on *our* specific way of interacting with the teaspoon or the balloon. Probability does not refer to the evolution of matter in itself. It relates to the evolution of those specific